4.9.2 Networking

4.9.2.1 Network topology

Content	Additional information
Understand: • physical star topology • logical bus network topology and: • differentiate between them • explain their operation	A network physically wired in star topology can behave logically as a bus network by using a bus protocol and appropriate physical switching.

4.9.2.2 Types of networking between hosts

Content	Additional information
Explain the following and describe situations where they might be used: • peer-to-peer networking • client-server networking.	In a peer-to-peer network, each computer has equal status. In a client-server network, most computers are nominated as clients and one or more as servers. The clients request services from the servers, which provide these services, for example file server, email server.

4.9.2.3 Wireless networking

Content	Additional information
Explain the purpose of WiFi.	A wireless local area network that is based on international standards. Used to enable devices to connect to a network wirelessly.
Be familiar with the components required for wireless networking.	Wireless network adapter. Wireless access point.
Be familiar with how wireless networks are secured.	Strong encryption of transmitted data using WPA (Wifi Protected Access)/WPA2, SSID (Service Set Identifier) broadcast disabled, MAC (Media Access Control) address allow list.
Explain the wireless protocol Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) with and without Request to Send/ Clear to Send (RTS/CTS).	Knowledge of Carrier Sense Multiple Access/ Collection Detection (CSMA/CD) as used in, for example, Ethernet, is not required.
Be familiar with the purpose of Service Set Identifier (SSID).	